

Module 2: FSx Assignment

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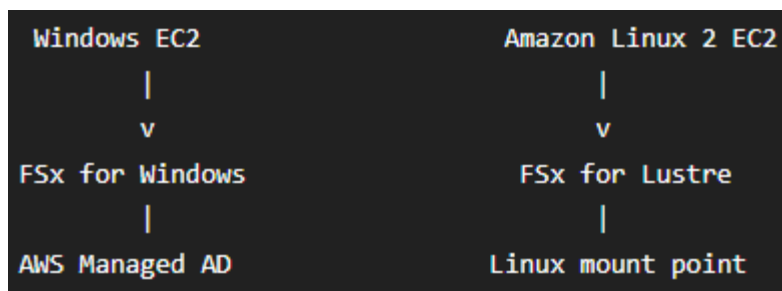
Problem Statement:

You work for XYZ Corporation and the current requirement in the organization is for faster file sharing, which can also help in data replication from on-premises infrastructure.

Tasks To Be Performed:

1. To Create an FSx file system for a Windows file server:
 - a. To make an AWS Managed Active Directory with a valid domain name
 - b. To connect it to Windows EC2 server
2. To create an FSx file system for Lustre and attach it to an Amazon Linux 2 instance.

Architecture



Part 1 — FSx for Windows File Server

Step 1 — Created AWS Managed Microsoft AD

- **AWS Console → Directory Service → Directories → Set up directory**
- Selected: **AWS Managed Microsoft AD**
- Directory DNS name: **mgib.intellipaat.edu**
- NetBIOS name: **EDU**
- Directory ID: **d-c3676c36b0**
- VPC: **vpc-0e9d79ad3a9956ee4**
- Zones: **eu-north-1a, eu-north-1b**
- DNS IP 1: **172.31.29.142**
- DNS IP 2: **172.31.32.139**
- AD Security Group: **sg-0a59969a446bac7b5**
- The directory administrator is: **Admin**, When using it from Windows, the username is: **EDU\admin**

The screenshot shows the AWS Directory Service console in the 'Set up a directory' wizard. The 'Enter directory information' step is active, showing the configuration for a new Microsoft AD directory. The directory type is 'Microsoft AD' and the operating system version is 'Windows Server 2019'. Two editions are available: 'Standard Edition' (selected) and 'Enterprise Edition'. The 'Standard Edition' is described as 'Best for small to medium sized businesses' and includes 1GB of storage for directory objects, optimized for up to 30,000 objects, and a cost of ~USD 89.2800/mo (USD 0.1240/hr)*. The 'Enterprise Edition' is described as 'Best for large businesses' and includes 17GB of storage for directory objects, optimized for up to 500,000 objects, and a cost of ~USD 296.6400/mo (USD 0.4120/hr)*. The 'Directory DNS name' is 'mgib.intellipaat.edu' and the 'Directory NetBIOS name - optional' is 'EDU'. The 'Directory description - optional' is 'Describe this directory'. The 'Admin password' is masked with dots. The console also shows a sidebar with navigation links for EC2, EFS, Directory Service, FSx, IAM, CloudWatch, S3, and Lambda.

Enter directory information

Directory information
A managed Microsoft Active Directory domain.

Directory type
Microsoft AD

Operating system version
Windows Server 2019

Edition
Microsoft AD is available in the following two editions:

- ☒ **Standard Edition**
Best for small to medium sized businesses.
 - 1GB of storage for directory objects
 - Optimized for up to 30,000 objects
 - ~USD 89.2800/mo (USD 0.1240/hr)*
 - * includes two domain controllers, USD 44.6400/mo for each additional domain controller.
- ☐ **Enterprise Edition**
Best for large businesses.
 - 17GB of storage for directory objects
 - Optimized for up to 500,000 objects
 - ~USD 296.6400/mo (USD 0.4120/hr)*
 - * includes two domain controllers, USD 148.3200/mo for each additional domain controller.

Directory DNS name
A fully qualified domain name. This name will resolve inside your VPC only. It does not need to be publicly resolvable.
mgib.intellipaat.edu

Directory NetBIOS name - optional
A short identifier for your domain. If you do not specify a NetBIOS name, it will default to the first part of your Directory DNS name.
EDU
Maximum of 15 characters, can't contain spaces or the following characters: \ / : * ? " < > | . It must not start with - or .

Directory description - optional
Descriptive text that appears on the details page after the directory has been created.
Describe this directory
Maximum of 128 characters, can only contain alphanumerics, and the following characters: _ @ # % * + = ? . / ! \ - . It may not start with a special character.

Admin password
The password for the default administrative user named Admin.

Passwords must be between 8 and 64 characters, not contain the word "admin", and include three of these four

Step 1

Step 2

Step 3

Step 4

Review & create

Choose VPC and subnets

Networking

The VPC that contains your directory. If you do not have a VPC with at least two subnets, you must create one.

VPC

vpc-0e9d79ad3a9956ee4 (172.31.0.0/16)

Create new VPC

Network type

IPv4 only

Subnets

subnet-088e08f0a01304957 (172.31.16.0/20, eu-north-1a)

subnet-01684b45c06381071 (172.31.32.0/20, eu-north-1b)

Create new subnet

Initial AD site name for this directory

Default-First-Site-Name

Cancel

Previous

Next

Directory Service

Active Directory

Directories

Directories shared with me

Your directory mgib.intellipat.edu (d-c3676c36b0) is being created! This can take up to 20-45 minutes.

See Detail

You can now extend your existing AD domain using AWS Managed Microsoft AD

With AWS Managed Microsoft AD, you can now create a hybrid directory to manage a single AD domain across your existing self-managed AD infrastructure and AWS. [Learn more](#)

Set up directory

Directories (1)

Find by directory ID or name

Directory ID	Directory name	Type	Edition	Multi-Region	Status	Launch date
d-c3676c36b0	mgib.intellipat.edu	Microsoft AD	Standard	Not applicable	Creating	Sep 24, 2026

Users

Groups

Networking & security

Scale & share

Application management

Maintenance

Networking details

Add IPv6 support

Availability zones

eu-north-1a

eu-north-1b

DNS IPv4 addresses

172.31.29.142

172.31.32.139

DNS IPv6 addresses

VPC

vpc-0e9d79ad3a9956ee4

Security group

sg-0a59969a446bac7b5

Subnets

subnet-088e08f0a01304957

subnet-01684b45c06381071

Status

Active

Last updated

Thursday, September 24, 2026

Launch time

Thursday, September 24, 2026

Step 2 — Created FSx for Windows File Server

FSx → File systems → Create file system

Selected: File system type: **FSx for Windows File Server**

Selected:

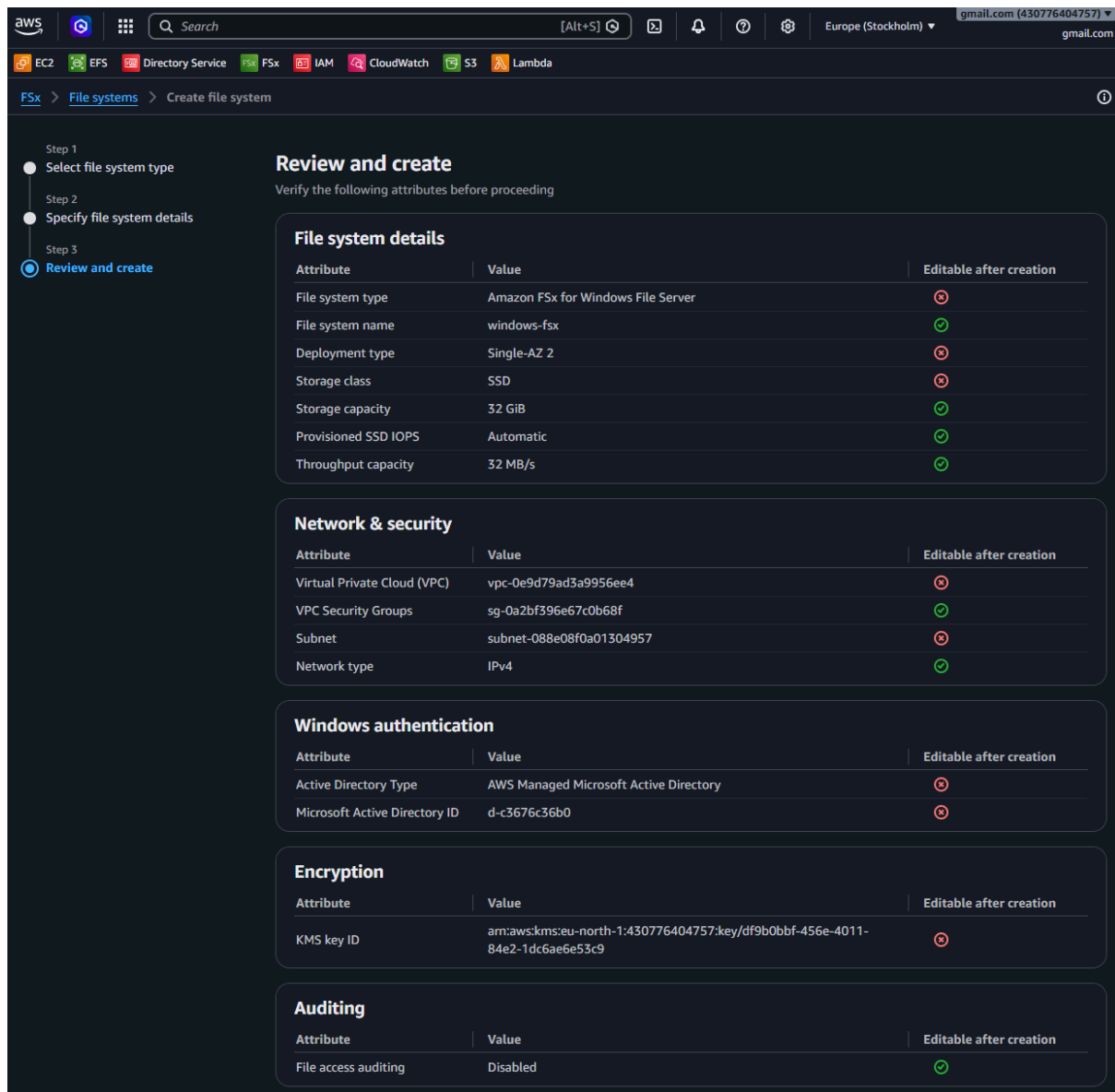
Name : **windows-fsx**
Deployment : **Single-AZ 2**
Storage class : **SSD**
Capacity : **32 GiB**
Joined the filesystem to : mgib.intellipat.edu
Used same VPC

The screenshot shows the AWS Management Console interface for creating a new FSx for Windows File System. The breadcrumb navigation at the top indicates the path: **FSx > File systems > Create file system**. On the left, a progress bar shows three steps: **Step 1: Select file system type**, **Step 2: Specify file system details** (which is the active step), and **Step 3: Review and create**.

The main section is titled **Specify file system details** and contains several configuration options:

- Creation method:** Two radio buttons are present. **Quick create** is selected, with a description: "Use recommended best-practice configurations. Most configuration options can be changed after the file system is created." The **Standard create** option is unselected, with a description: "You set all of the configuration options, including specifying performance, networking, security, backups, and maintenance."
- Quick configuration:** This section contains several fields and options:
 - File system name - optional:** A text input field containing "windows-fsx". A note below states: "Maximum of 256 Unicode letters, whitespace, and numbers, plus + - = . _ : /".
 - Deployment type:** Three radio buttons are shown. **Single-AZ 2** is selected. Descriptions for each option are provided: "Multi-AZ (Recommended)" for production workloads, "Single-AZ 2" as the latest generation supporting SSD and HDD, and "Single-AZ 1".
 - SSD storage capacity:** A text input field set to "32" GiB. A note below states: "Minimum 32 GiB; Maximum 65,536 GiB".
 - Virtual Private Cloud (VPC):** A dropdown menu showing "vpc-0e9d79ad3a9956ee4 (CIDR: 172.31.0.0/16)".
 - Subnet:** A dropdown menu showing "subnet-088e08f0a01304957 (eu-north-1a | eun1-az1)".
 - Windows authentication:** Two radio buttons are shown. **AWS Managed Microsoft Active Directory** is selected. A description states: "Choose an Active Directory to provide user authentication and access control for your file system." The **Self-managed Microsoft Active Directory** option is unselected.
 - AWS Managed Microsoft Active Directory:** A text input field contains "mgib.intellipat.edu | d-c3676c36b0". A "Create new directory" link is provided next to it.

At the bottom right of the console, there are three buttons: **Cancel**, **Back** (highlighted with a blue border), and **Next** (highlighted in orange).



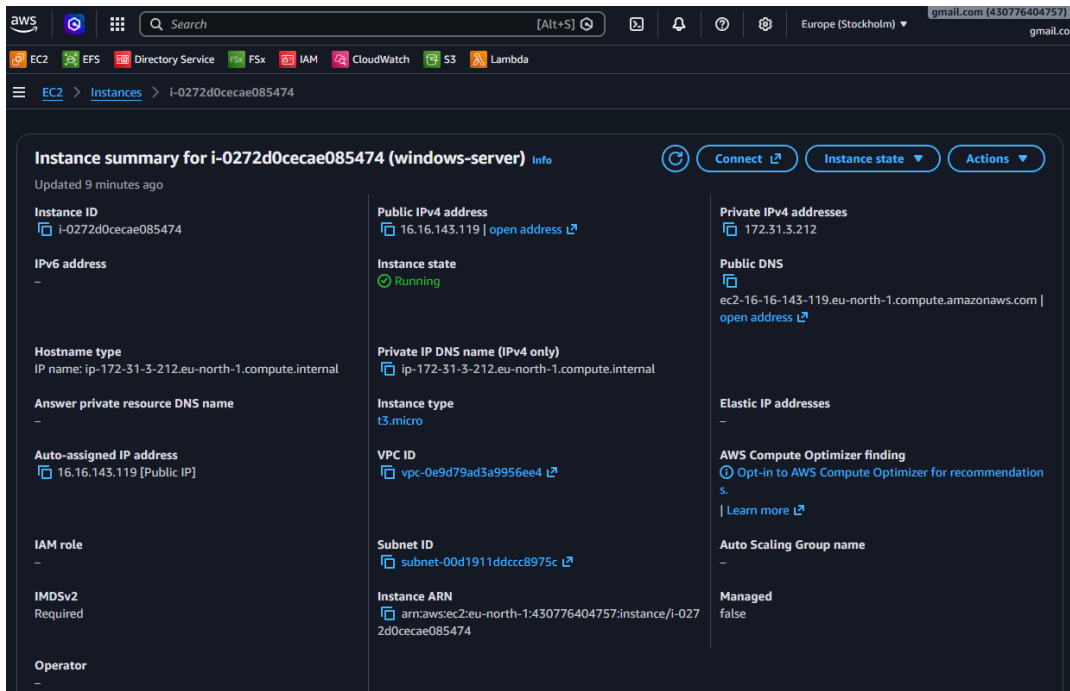
Step 3 — Launched and connected Windows EC2

1. Launched instance

EC2 → Launch instance

Chooosed: Windows_Server-2025-English-Full-Base-2026.09.17

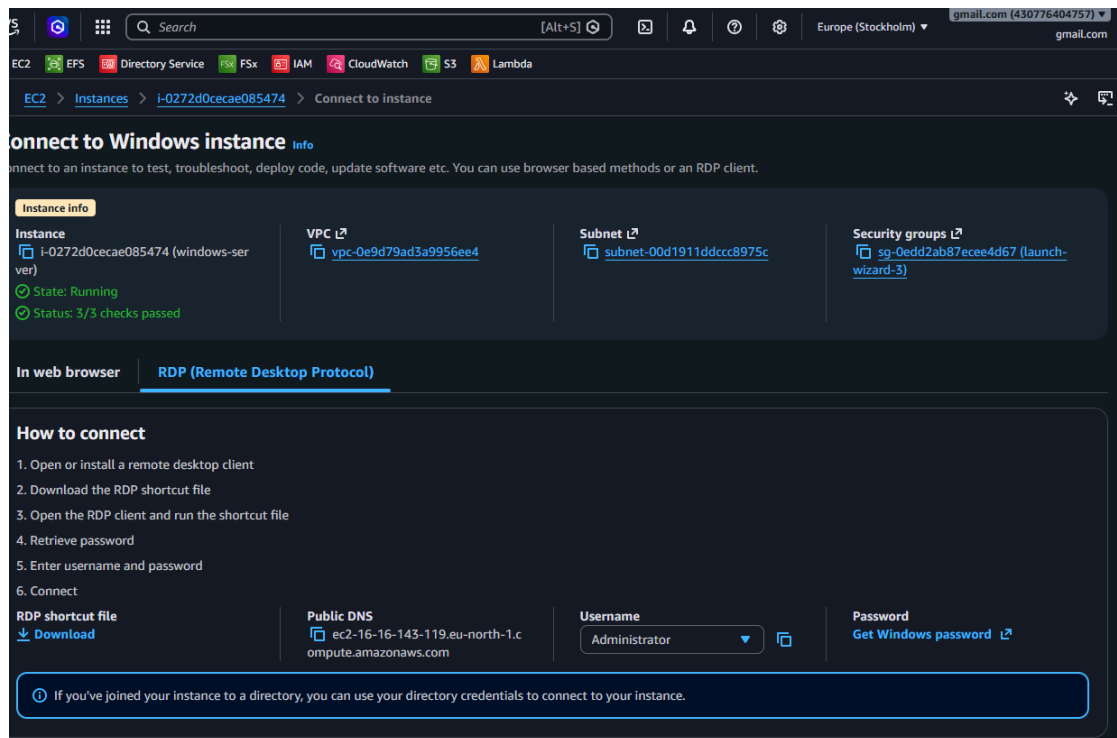
Used the **same VPC** as the Managed AD.



Details	Status and alarms	Monitoring	Security	Networking	Storage	Tags
VPC ID vpc-0e9d79ad3a9956ee4 ↗		Subnet ID subnet-00d1911ddccc8975c ↗		Availability zone ↗ eu-north-1c		
Availability zone ID ↗ eun1-az3		Outpost ID -				
▼ IP addresses Info						
Public IPv4 address ↗ 16.16.143.119 open address ↗		Private IPv4 addresses ↗ 172.31.3.212		IPv6 addresses -		
Secondary private IPv4 addresses -		Carrier IP addresses (ephemeral) -				
▼ Hostname and DNS Info						
Public DNS ↗ ec2-16-16-143-119.eu-north-1.compute.amazonaws.com open address ↗		Private IP DNS name (IPv4 only) ↗ ip-172-31-3-212.eu-north-1.compute.internal		IPv4-only IP based name: A record only ↗ ec2-16-16-143-119.eu-north-1.compute.amazonaws.com		
Dualstack - IP based name: A and AAAA record -		IPv6-only - IP based name: AAAA record only -		Public hostname type public-ipv4-dns-name		
Private hostname type IP name: ip-172-31-3-212.eu-north-1.compute.internal		Use RBN as guest OS hostname ↗ Disabled		Answer RBN DNS hostname IPv4 ↗ Disabled		
Answer RBN DNS hostname IPv6 -		Answer private resource DNS name -				

2. Connected to windows EC2

- EC2** → **Instances** → Select **windows-server EC2** → Clicked **Connect** → **RDP client**
- Got Windows administrator password using your key pair.
- Download/open the RDP file. Connected to the Windows server.



Step 4 — Joined the Windows EC2 to the domain

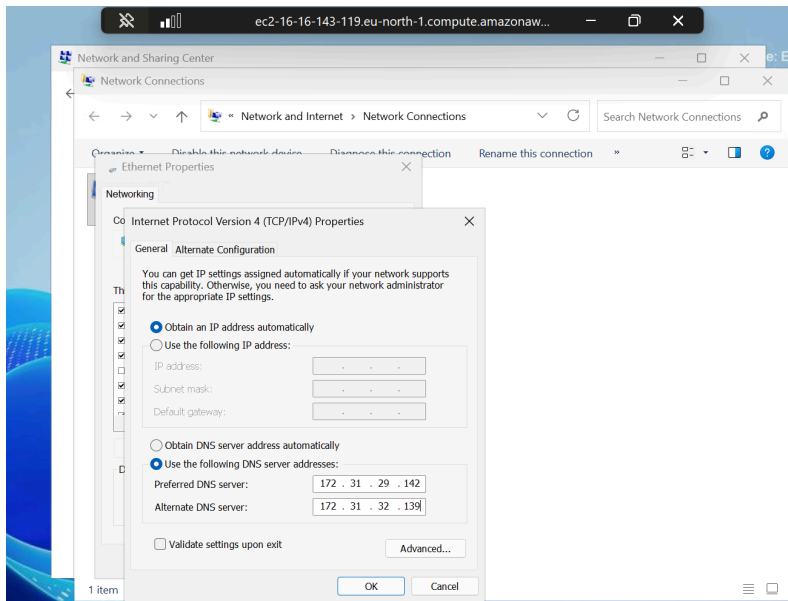
1. Set Windows DNS

Control Panel → Network and Internet → Network Connections
Right-clicked network adapter → **Properties** → **IPv4** → **Properties**.

Set DNS to:

Preferred DNS:
172.31.29.142

Alternate DNS:
172.31.32.139



```

Administrator: Command Prompt
ec2-16-16-143-119.eu-north-1.compute.amazonaws...

Microsoft Windows [Version 10.0.26100.33451]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\Administrator>nslookup mgib.intellipat.edu
Server: ip-c613016f.mgib.intellipat.edu
Address: 172.31.29.142

Name: mgib.intellipat.edu
Addresses: 172.31.29.142
           172.31.32.139

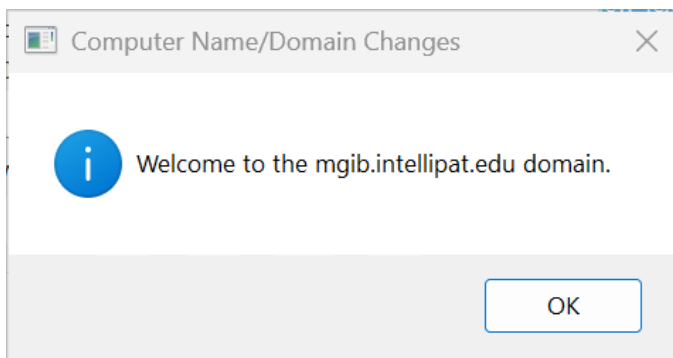
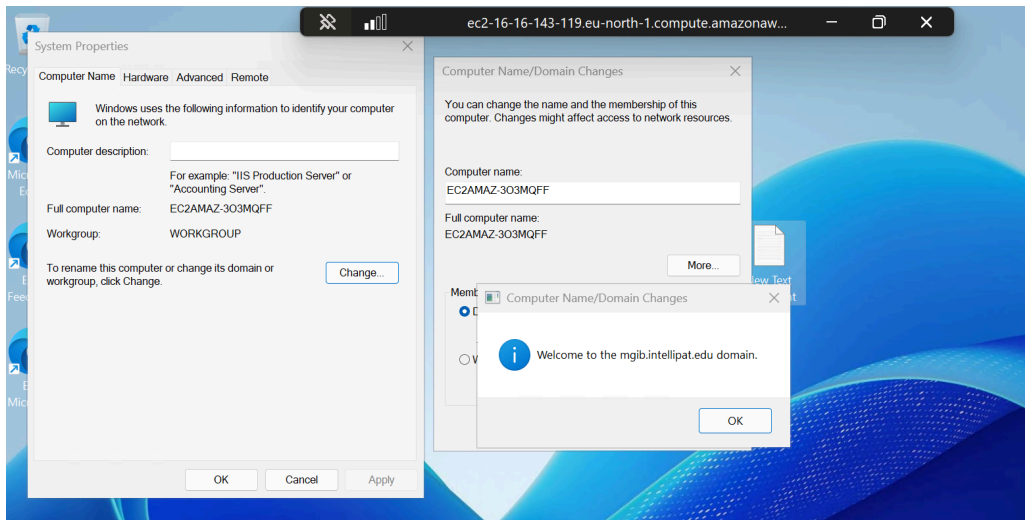
C:\Users\Administrator>

```

2. Join Windows EC2 to Active Directory

Windows: **System** → **Advanced system settings** → **Computer Name** → **Change**

- Selected: **Domain**
- Enter: **mgib.intellipat.edu**
- credentials:
 - Username: **EDU\admin**
 - Password: **<AD Admin password>**
- **Windows EC2 successfully joined the domain.**
- **Restart Windows**



3. Map the Windows FSx share

PowerShell:

net use Z: \\amznfsxoojuuqdv.mgib.intellipaat.edu\share

Username: **EDU\admin**

Password: <AD Admin password>

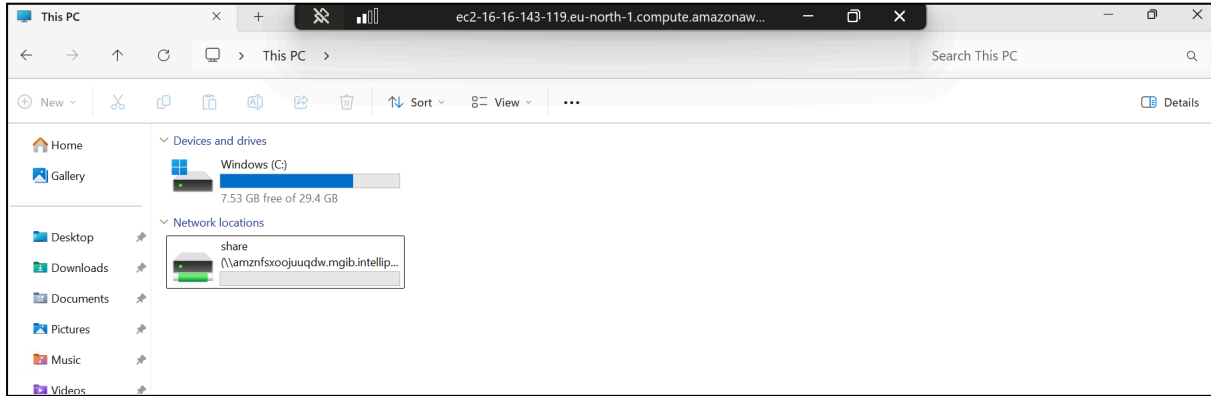
It returned:

The command completed successfully.

✅ The FSx share was successfully mapped to:

```
Administrator: Windows Powe... ec2-16-16-143-119.eu-north-1.compute.amazonaw...
PS C:\Windows\system32> net use Z: \\amznfsxoojuuqdw.mgib.intellipat.edu\share
Enter the user name for 'amznfsxoojuuqdw.mgib.intellipat.edu': EDU\admin
Enter the password for amznfsxoojuuqdw.mgib.intellipat.edu:
The command completed successfully.

PS C:\Windows\system32>
```



Part 2 — FSx for Lustre

1. Created FSx for Lustre

AWS Console → FSx → File systems → Create file system

Selected:

File system type	: FSx for Lustre
Deployment class	: Scratch, SSD
EFA	: Disabled
Storage capacity	: 1.2 TB
Throughput	: 200 MB/s/TiB
Data compression	: NONE
Lustre version	: 2.15

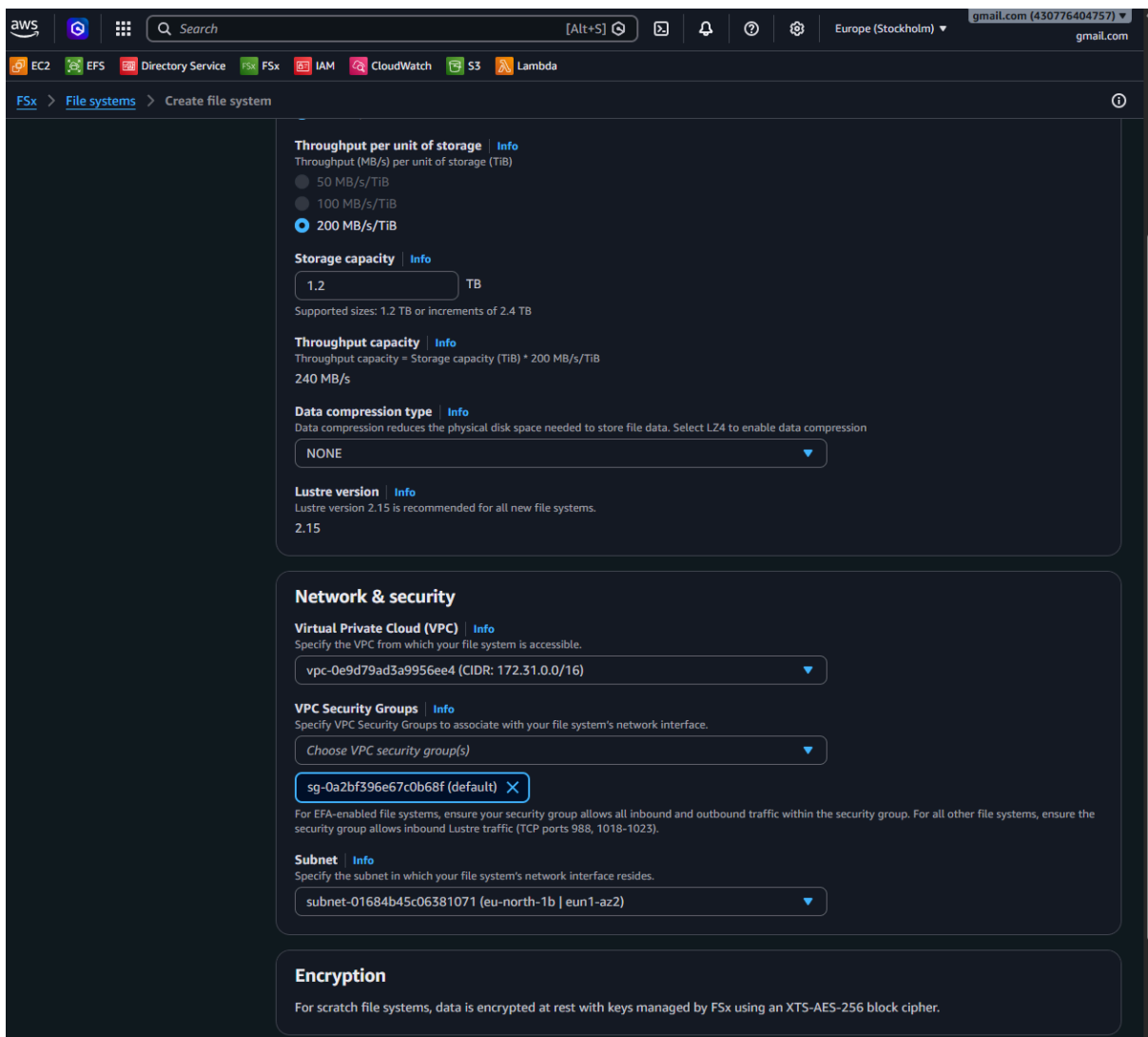
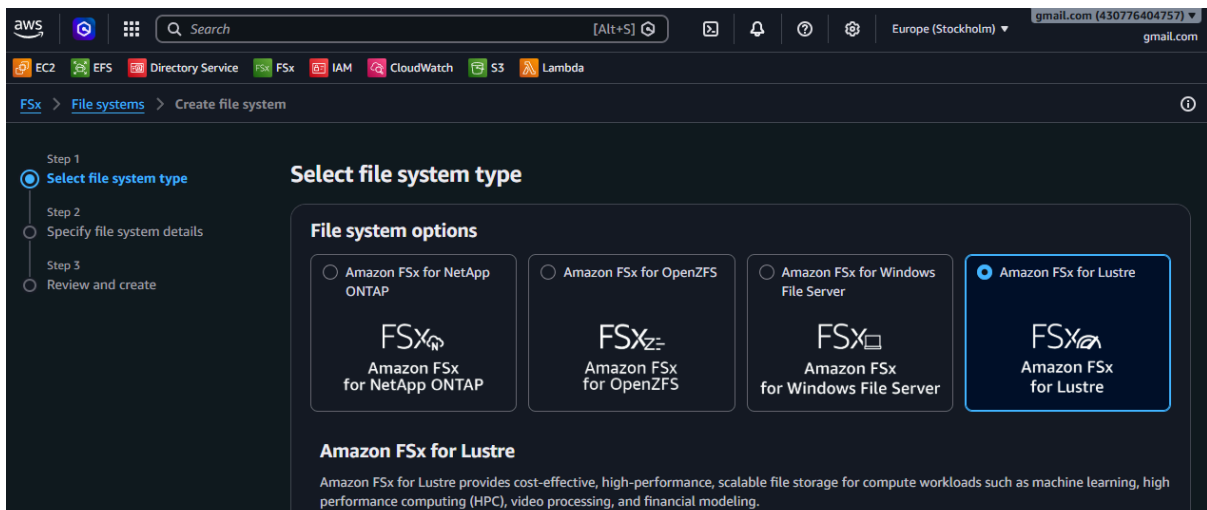
Network

VPC: vpc-0e9d79ad3a9956ee4

Subnet: subnet-01684b45c06381071

Choose the FSx security group.

Create the filesystem and wait for: Status: Available



2. Launched Linux EC2

EC2 → Launch instance

Used:

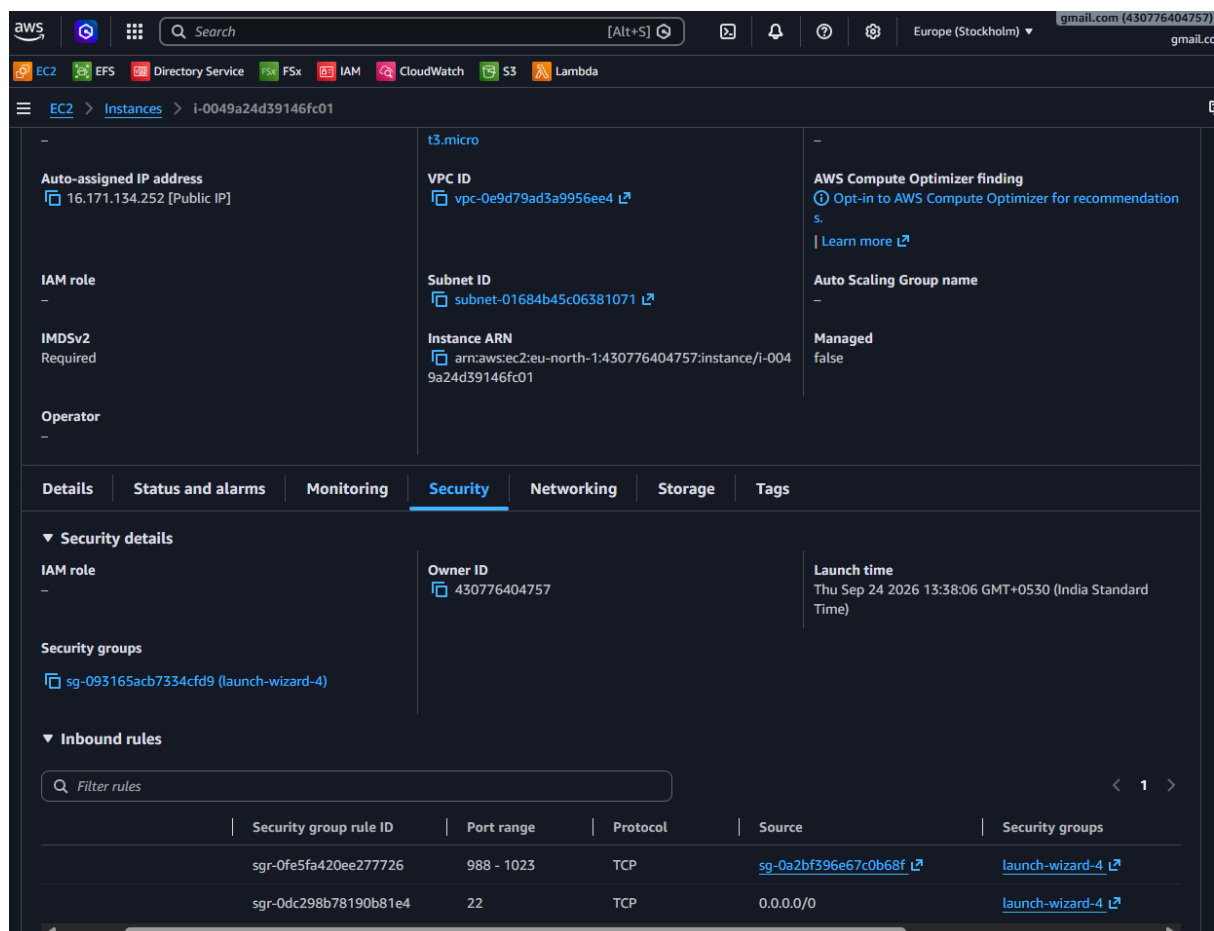
AMI: Amazon Linux 2023

Instance type: t3.micro

VPC: same VPC as FSx

Subnet: same/subnet environment

Public IP: Enable



3. Configure Lustre Security Groups

For **FSx security group**, allowed:

Custom TCP 988 Source: Linux EC2 security group

Custom TCP 1018-1023 Source: Linux EC2 security group

For the **Linux EC2 security group**, allow the corresponding Lustre traffic from the FSx security group.

Description - required | [Info](#)

launch-wizard-4 created 2026-09-24T07:58:14.424Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0) [Remove](#)

Type	Info	Protocol	Info	Port range	Info
ssh		TCP		22	

Source type | [Info](#)

Anywhere

Source | [Info](#)

Q Add CIDR, prefix list or security

0.0.0.0/0 X

Description - optional | [Info](#)

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 988-1023) [Remove](#)

Type	Info	Protocol	Info	Port range	Info
Custom TCP		TCP		988-1023	

Source type | [Info](#)

Custom

Source | [Info](#)

Q Add CIDR, prefix list or security

Description - optional | [Info](#)

e.g. SSH for admin desktop

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. X

4. Mounted FSx Luster in Linux EC2

Install the Lustre client

Ran in linux EC2: `sudo dnf install -y lustre-client`

```

aws
Q Search [Alt+S]
Europe (Stockholm) gmail.com (430776404757) gmail.com

EC2 EFS Directory Service FSx IAM CloudWatch S3 Lambda

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Thu Sep 24 08:11:35 2026 from 13.48.4.202
[ec2-user@ip-172-31-46-36 ~]$ sudo dnf install -y lustre-client
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
731 kB/s | 82 kB 00:00

Package Architecture Version Repository Size
Installing:
lustre-client x86_64 2.15.6-32.amzn2023 amazonlinux 755 k

Transaction Summary
Install 1 Package

Total download size: 755 k
Installed size: 2.4 M
Downloading Packages:
lustre-client-2.15.6-32.amzn2023.x86_64.rpm 18 MB/s | 755 kB 00:00
Total 7.9 MB/s | 755 kB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : lustre-client-2.15.6-32.amzn2023.x86_64 1/1
Installing : lustre-client-2.15.6-32.amzn2023.x86_64 1/1
Running scriptlet: lustre-client-2.15.6-32.amzn2023.x86_64 1/1
Verifying : lustre-client-2.15.6-32.amzn2023.x86_64 1/1

Installed:
lustre-client-2.15.6-32.amzn2023.x86_64

Complete!

```

Created the mount directory

FSx → your Lustre filesystem → Attach

Used the **exact command from the AWS Attach page**:

```
sudo mkdir -p /fsx
```

```
sudo mount -t lustre -o relatime,flock
```

```
fs-09668d991523657d9.fsx.eu-north-1.amazonaws.com@tcp:/jbe3rbev /fsx
```

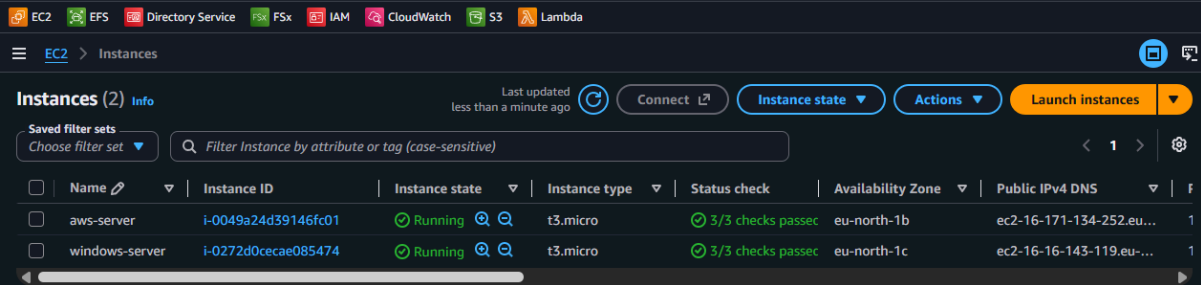
```
[ec2-user@ip-172-31-46-36 ~]$ sudo mkdir -p /fsx
[ec2-user@ip-172-31-46-36 ~]$ sudo mount -t lustre -o relatime,flock fs-09668d991523657d9.fsx.eu-north-1.amazonaws.com@tcp:/jbe3rbev /fsx
[ec2-user@ip-172-31-46-36 ~]$
[ec2-user@ip-172-31-46-36 ~]$ mount | grep lustre
fs-09668d991523657d9.fsx.eu-north-1.amazonaws.com@tcp:/jbe3rbev on /fsx type lustre (rw,relatime,flock)
[ec2-user@ip-172-31-46-36 ~]$ df -h /fsx
Filesystem                                Size  Used Avail Use% Mounted on
fs-09668d991523657d9.fsx.eu-north-1.amazonaws.com@tcp:/jbe3rbev 1.2T   24K  1.2T   1% /fsx
[ec2-user@ip-172-31-46-36 ~]$ echo "FSx for Lustre test successful" | sudo tee /fsx/test.txt
FSx for Lustre test successful
```

Result Summary:

Successfully created and configured **two Amazon FSx file systems**:

1. **FSx for Windows File Server** – integrated with AWS Managed Microsoft AD and successfully mapped to the Windows EC2 instance as a network drive (Z:).
2. **FSx for Lustre** – configured with Amazon Linux 2023 EC2, mounted successfully, and verified by creating and reading a test file.

Both FSx file systems were successfully accessed and tested from their respective EC2 instances.



	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	
☐	aws-server	i-0049a24d39146fc01	Running	t3.micro	3/3 checks passed	eu-north-1b	ec2-16-171-134-252.eu...	1
☐	windows-server	i-0272d0cecae085474	Running	t3.micro	3/3 checks passed	eu-north-1c	ec2-16-16-143-119.eu...	1